

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0702

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Debugging & Optimisation Task

Code of Conduct:

*I K. Yamini (192572181) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

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RAH1

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
 - Determining the network prefix of each device.
 - Verifying whether both devices belong to the same subnet.
 - Identifying the addressing or prefix mismatch.
 - Optimizing the IPv6 configuration by assigning appropriate addresses.
 - Calculating the number of available host addresses within the corrected subnet.
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3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

COR - TOOL - 3Debugging & Optimization1. IPv4 configuration Debugging1. Problem Identification

The given IP address is 192.168.10.65/26.

subnet mask 255.255.255.192 = /26, which creates subnets of size 64.

subnets :

- 192.168.10.0 - 63
- 192.168.10.64 - 127
- 192.168.10.128 - 191
- 192.168.10.192 - 255

IP 192.168.10.65 belongs to subnet 192.168.10.64/26.

2. Debugging

- Network Address : 192.168.10.64
- Broadcast Address : 192.168.10.127
- Valid Host Range : 192.168.10.65 - 192.168.10.126

check gateway :

- Given gateway = 192.168.10.1
- This belongs to 192.168.10.0/26 subnet
- so gateway is incorrect.

3. Error Identification

- Gateway is in different subnet.
- Hence communication fails.

4. optimization

correct configuration:

- IP Address : 192.168.10.66
- subnet mask : 255.255.255.192
- Default gateway : 192.168.10.65

5. Analysis

- devices must be in same subnet.
- gateway must belong to same network range.

6. usable Hosts calculation

Formula :

$$\text{Usable hosts} = 2^{(32-8)} - 2 = 64 - 2 = 62$$

2. IPv6 Debugging

1. Problem Identification

Device A : 2001 : db8 : :: ff00 : 42 : 8329 / 64

Device B : 2001 : db8 : 0 : 0 : 1 : :: 1 / 64

2. Debugging

Expanded :

• Device A :

2001 : db8 : 0000 : 0000 : 0000 : ff00 : 0042 : 8329

• Device B :

2001 : db8 : 0000 : 0000 : 0001 : 0000 : 0000 : 0001

3. Network Prefix

• Device A prefix : 2001 : db8 : 0 : 0 : :: / 64

• Device B prefix : 2001 : db8 : 0 : 0 : 1 : :: / 64

4. Error Identification

- prefix mismatch $\rightarrow$ different subnets
- Hence no communication

5. Optimization

Assign same subject:

Example:

- Device A : $2001 : db8 : 0 : 0 :: 1/64$
- Device B : $2001 : db8 : 0 : 0 :: 2/64$

6. Analysis

IPv6 devices must have:

- same 64-bit prefix
- only host part should differ

7. Host calculation

IPv6 hosts = 2^{64} (very large)

3. Subnet Optimization (VLSM)

1. Problem Identification

Network : $192.168.1.0/24$ divided into $/26, /27, /28$

Issue:

- some subnets $\rightarrow$ insufficient IPs
- some $\rightarrow$ unused IPs

2. Debugging

<u>subnet</u>	<u>Range</u>	<u>Usable Hosts</u>
/26	64 IPs	62
/27	32 IPs	30
/28	16 IPs	14

3. Error Identification

- Large departments assigned small subnets.
- small departments assigned large subnets
→ inefficient allocation.

4. Optimization (VLSM)

steps:

1. Arrange departments by size.
2. Assign largest subnet first

Example:

- Dept A → /26
- Dept B → /27
- Dept C → /28

5. Analysis

- VLSM reduces IP wastage
- Improves network efficiency or efficiency.

4. IPv6 Debugging

1. Problem Identification

Two devices are configured with IPv6 addresses:

- Device A : 2001:db8::ff00:42:8329/64
- Device B : 2001:db8:0:0:1::/64

Even though they are connected to the same network, communication fails.

6. Debugging

Expansion of IPv6 Addresses

• Device A :

2001 : 0db8 : 0000 : 0000 : 0000 : 1f00 : 0042 : 8329

• Device B :

2001 : 0db8 : 0000 : 0000 : 0001 : 0000 : 0000 : 0001

3. Network prefix Identification

• Device A prefix : 2001 : db8 : 0 : 0 :: /64

• Device B prefix : 2001 : db8 : 0 : 0 : 1 :: /64

4. Error Identification

- Device A and Device B belong to different subnets.
- Prefix mismatch occurs at the 5th block.
- Hence, device cannot communicate.

5. Optimization

correct configuration

Assign both devices in same subnet :

• Device A : 2001 : db8 : 0 : 0 :: /64

• Device B : 2001 : db8 : 0 : 0 :: 2/64

6. Analysis

- IPv6 communication requires same 64-bit network prefix.
- only host portion should differ.
- Prefix mismatch leads to routing failure.

6. Host Address calculation

- IPv6 /64 subnet supports :
 2^{64} host addresses
- This provides extremely large address

5. Routing Table Debugging

1. Problem Identification

Routing table:

- 192.168.1.0/24 → 10.0.0.2
- 192.168.2.0/24 → 10.0.0.3
- Default → 10.0.0.1

Destination: 192.168.2.100

2. Debugging

- matches 192.168.2.0/24
- Next hop = 10.0.0.3

3. Error Identification

Possible issue:

- wrong next hop
- Router misconfiguration.
- Interface down.

4. Optimization

- verify next hop connectivity
- Add specific route if needed
- Ensure interface active.

5. Analysis

- Longest prefix match selects correct route
- If no match → default route used

6. Default Route

- used when no match exists
- Next hop: 10.0.0.1